HW

1. $F(s) = \dfrac{s+7}{s^2+4s+8}$ 2. $f(s) = \dfrac{3s-137}{s^2+2s+401}$

3 $7/(s-1)^3$ 4. $\sqrt{8}/(s+\sqrt{2})^3$ 5 $\dfrac{15}{s^2+4s+29}$

6 $\pi/(s+\pi)^2$ 7. $s-6/(s-1)^2+4$

8. $4s-2/s^2-6s+18$ 9. $2s-56/s^2-4s-12$

1. $\dfrac{s+7}{s^2+4s+4+4} = \dfrac{s+7}{(s+2)^2+4} = \dfrac{s+2}{(s+2)^2+4} + \dfrac{5}{(s+2)^2+4}$

$= e^{-2t} L^{-1}\left\{ \dfrac{s}{s^2+4} + \dfrac{5}{2} \dfrac{2}{s^2+2^2} \right\}$

$= e^{-2t}\left(\cos 2t + \frac{5}{2} \sin 2t\right)$

2 $\dfrac{3s-137}{(s+1)^2+400}$

$\boxed{3\left(\dfrac{s}{(s+1)^2+400}\right) - \dfrac{137}{20}\left(\dfrac{20}{(s+1)^2+400}\right)}$

$3\left(\dfrac{s+1}{(s+1)^2+400} - \dfrac{1}{20}\dfrac{20}{(s+1)^2+400}\right) - \dfrac{137}{20}\dfrac{20}{(s+1)^2+400}$

$e^{-t}\left[3\left(\cos 20t - \frac{1}{20}\sin 20t\right) - \frac{137}{20}\sin 20t\right]$

3. $7/(s-1)^3 = e^{t} * (1/s^3)$

$= 7e^{t} \cdot \frac{t^2}{2}$

4. $\sqrt{8}/(s+\sqrt{2})^3 = e^{-\sqrt{2}} (\sqrt{8}/s^3)$

$= e^{-\sqrt{2}} \frac{\sqrt{8}}{2} (2/s^3)$

$= e^{-\sqrt{2}} \sqrt{8}/2 \; t^2.$

5. $15/s^2+4s+29$

$3 \cdot e^{-2t} \sin 5t$

6. $\pi/(s+\pi)^2 \cdot \quad e^{-\pi t} \; \pi t$

$4s-2+10-10$

$4s-12+10$

$4(s-3)$

7. $s-6/(s-1)^2+4.$

$\frac{s-1}{(s-1)^2+4} - 5\left(\frac{1}{(s-1)^2+4}\right) = e^{t}\left[\cos 2t - \frac{5}{2}\sin 2t\right]$

8. $\frac{4s-2}{s^2-6s+18} = \frac{4s-2}{(s-3)^2+9} = 4\frac{(s-3)}{(s-3)^2+3^2} + \frac{10}{(s-3)^2+3^2}.$

$= e^{+3t}\left[(4\cos 3t)+10\sin 3t\right]$

9. $\frac{2s-56}{s^2-4s-12} \quad \frac{2s-56}{s^2-4s-16+4} \quad \frac{2(s-28)}{(s-2)^2-16}$

$= 2\left[\frac{s-2}{(s-2)^2-16} - \frac{26}{(s-2)^2-16}\right]$

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$2\left[\frac{s}{s^2-16} - \frac{26}{4}\frac{4}{s^2-16}\right]$

$2e^{2t}\left(\cosh 4t - 13/2 \sinh 4t\right).$